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Seeing Motion Through Polarity for Event-based Action Recognition

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Abstract

Event-based Action Recognition (EAR) provides a promising pathway for understanding dynamic behaviors under challenging conditions. Recent progress in vision-language models has introduced a cross-modal learning paradigm into EAR, enabling models to associate event streams with textual semantics for enhancing conceptual understanding. However, existing methods typically overlook the intrinsic polarity-driven motion cues that are fundamental to event data, leading to suboptimal spatiotemporal representations. To address this limitation, we propose a POlarity Knowledge Enhanced framework (POKER), which explicitly incorporates event polarity-aware motion knowledge across visual and textual modalities. POKER consists of two synergistic components: Polarity Motion Capturer (PMC) and Polarity Motion Reasoner (PMR). Specifically, PMC decouples positive and negative polarities to capture polarity-sensitive motion cues, while PMR semantically analyzes polarity-induced motion dynamics via large language models. Through the polarity alignment, POKER couples semantic reasoning with visual dynamics, achieving more discriminative representations. Extensive experiments on multiple benchmarks demonstrate that POKER enhances performance across diverse event representations.

1. Introduction

Human action recognition is a fundamental task in computer vision, supporting applications such as intelligent surveillance [38, 39, 44] and human–computer interaction [29, 43]. RGB-based methods primarily rely on RGB data to capture spatio-temporal dynamics [12, 13, 27]. However, these methods rely on frame-based imaging mechanisms with limited motion capture capability, often failing in scenes involving rapid movement or significant lighting changes. Furthermore, the high redundancy of RGB data increases computational costs, simultaneously raising concerns about privacy leakage. Event cameras, as a new type of bio-inspired vision sensor, asynchronously record per-pixel brightness changes to generate sparse event streams [9, 18, 20, 36]. Compared with conventional RGB cameras, event cameras offer unique properties such as a high dynamic range, low power consumption, and inherent sparsity, enabling robust visual perception under extreme lighting, high-speed motion, and privacy-sensitive conditions. Benefiting from these unique properties, researchers have begun exploring the potential of event data for dynamic scene understanding, giving rise to the emerging field of Event-based Action Recognition (EAR) [22, 25, 32–34, 42, 47, 52].

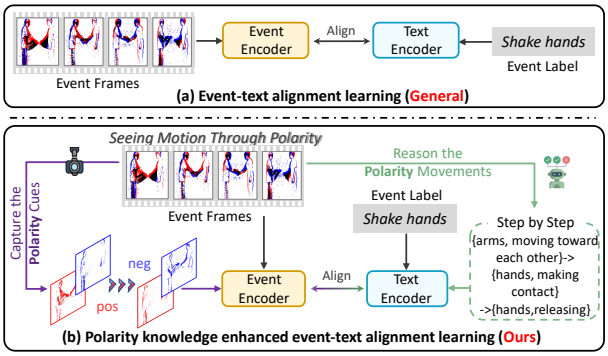


Figure 1. Insight of the proposed POKER. (a) Previous event-text alignment methods adhere to general paradigms, which limit the sufficiency of cross-modal semantic alignment between events and text. (b) Our approach explicitly leverages the polarity-aware motion knowledge embedded in event streams from both visual and textual modalities, thereby achieving significantly enhanced event-text alignment.

EAR aims to exploit the dynamic cues embedded in event streams for robust motion understanding, offering a novel paradigm for intelligent perception in dynamic environments. However, a key challenge in EAR lies in effectively leveraging the spatially sparse and temporally asynchronous nature of event data. Existing studies can be broadly categorized based on the fundamental event rep-

resentations: (1) spatiotemporal-preserved representations, such as event points or event voxels [10, 17]; and (2) frame-stacked representations, notably event frames or time surfaces [26, 47]. The former typically employs point-based encoders (*e.g.*, GCNs or SNNs). While these architectures are intrinsically compatible with the asynchronous event characteristics and naturally offer low power consumption, the recognition performance often remains suboptimal compared to frame-based methods. Conversely, the latter leverages powerful frame-based encoders (*e.g.*, CNNs or Transformers). By converting sparse events into dense frame representations, this approach achieves superior performance and has therefore attracted considerable research attention.

In recent years, the rapid advancement of vision-language models has driven EAR research toward cross-modal learning paradigms, aiming to improve the semantic robustness of action recognition. As shown in Fig. 1 (a), conventional frameworks establish robust event-text associations through cross-modal alignment learning. Despite these advances, a key limitation remains. Existing cross-modal methods typically treat event streams as stacked frames, ignoring the unique motion dynamics inherent to the events. This simplification neglects the directional and temporal information encoded in event polarities, which are critical for understanding motion cues. Consequently, semantic alignment between modalities remains incomplete, resulting in features with limited discriminative power.

To tackle the limitation in previous cross-modal methods, we anchor our solution in the intrinsic mechanism of event generation. An event is triggered when the logarithmic brightness change at a pixel reaches a threshold, where the direction of the change is encoded by its polarity (positive for brightening, negative for darkening) [3, 6, 48]. From a motion perspective, positive and negative polarities capture the leading and trailing edges of motion, reflecting the directional evolution of dynamic scenes. Building on this decisive principle, we seek to understand the motion cues encapsulated within the polarity. As shown in Fig. 1 (b), we aim to collaboratively extract and integrate polarity-aware motion cues from both the visual and textual modalities, thus strengthening the event-text learning framework.

Formally, we introduce the POlarity Knowledge Enhanced framework (POKER) to fully exploit polarity-aware motion information inherent in the event stream. Specifically, POKER comprises two synergistic components: Polarity Motion Capturer (PMC) and Polarity Motion Reasoner (PMR). The PMC is designed to decompose the event stream by polarity, capturing temporal correlations within each polarity and spatial correlations across polarities, thereby comprehensively modeling the spatiotemporal motion dynamics in the event data. Meanwhile, the PMR guides multi-modal large language models to reason about motion changes induced by key polarities through a progressive reasoning prompt, generating semantic descriptions with polarity awareness. We perform polarity alignment between polarity-enhanced visual and textual features, effectively integrating visual dynamics with semantic reasoning to enable more discriminative feature learning.

The main contributions are summarized as follows,

- **New polarity-aware framework:** We propose a Polarity Knowledge Enhanced framework (POKER) for event-based action recognition, which integrates polarity-driven motion knowledge from both textual and visual modalities to explicitly enrich the spatiotemporal representations and improve feature discriminability.
- **Dual-polarity synergy:** We design two synergistic modules, Polarity Motion Capturer and Polarity Motion Reasoner, which collaboratively exploit cross-modal polarity cues and couple semantic reasoning with visual dynamics via a polarity alignment.
- **Comprehensive evaluation:** Extensive experiments on multiple datasets validate the effectiveness and scalability of our framework, consistently enhancing recognition performance across diverse event representations.

2. Related Work

2.1. Single-modal Event-based Action Recognition

Single-modal methods for EAR are primarily categorized based on the format of event data representation. One major stream of work aims to preserve the asynchronous nature of event data, representing it as event points or event voxels to capture fine-grained spatiotemporal relationships. These approaches typically employ bio-inspired Spiking Neural Networks [4, 34] or specialized Graph Convolutional Networks [8, 41]. For instance, SpikePoint [24] proposes an end-to-end SNN architecture that concurrently extracts global and local features to effectively process sparse event streams. Similarly, VMv-GCN [41] introduces a novel volumetric multi-view framework designed to extract critical spatial and temporal information from event voxels. Concurrently, another significant stream of work chooses to aggregate the asynchronous event streams into regular tensors [16, 28, 37], subsequently applying powerful architectures such as Vision Transformers or Mamba-based models. For example, EventTransAct [5] utilizes a video transformer network and incorporates specific enhancements to handle event frames. Furthermore, PAST-SSM [50] learns complex spatiotemporal dependencies from encoded event frame features using a state space model.

2.2. Cross-modal Event-based Action Recognition

The limited ability of single-modal methods to extract sufficient information from sparse event streams has driven recent EAR research to shift towards multi-modal and cross-modal paradigms. Early multi-modal efforts, such as TSC-

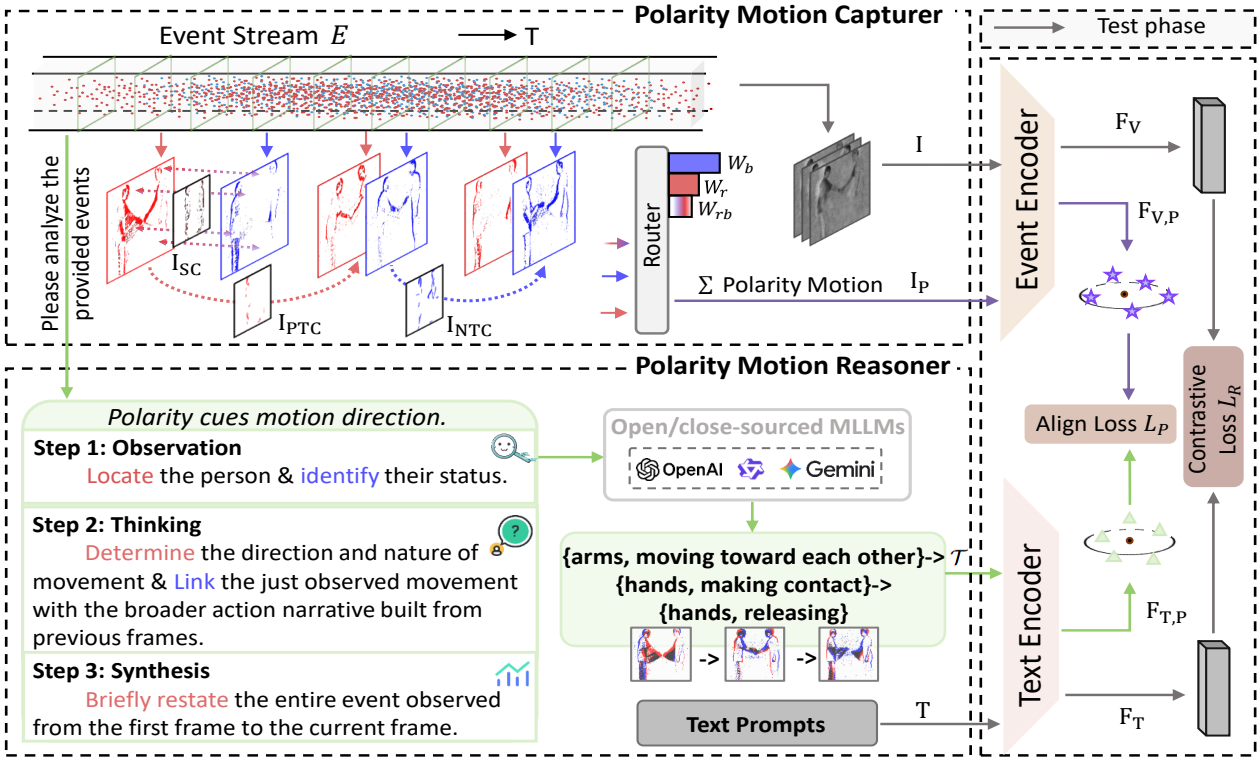


Figure 2. The overall framework of our proposed POKER enhances feature discriminability via polarity-aware motion knowledge. It consists of event and text encoders augmented by Polarity Motion Capturer (PMC) and Polarity Motion Reasoner (PMR). During training, the encoders process two distinct data streams: one for generating base features from the original event frames and text prompts, and another for generating polarity-enhanced features from the PMC's decoupled visual streams and the PMR's generated reasoning descriptions. These base and enhanced features are jointly optimized through contrastive learning and a polarity alignment loss to achieve robust cross-modal coupling. Critically, inference relies solely on the base features for final action prediction.

Former [34], focus on fusing event data with RGB frames to enhance perception for event data. More recently, the success of Vision-Language Models has accelerated cross-modal knowledge transfer, leveraging rich language semantics to address the high semantic uncertainty in event data. Representative examples include ExACT [52], which pioneers the use of language semantics for event frame processing, and EZSR [45] and EventGPT [19], which adapt VLM knowledge for scalable zero-shot recognition and MLLM-based event stream comprehension, respectively. However, a persistent limitation is that these works often overlook the intrinsic polarity clues fundamentally embedded within the event generation mechanism, thereby restricting the integrity of semantic alignment. In this work, we explicitly leverage event-intrinsic polarity knowledge, achieving semantically robust cross-modal alignment.

2.3. Knowledge Enhanced Methods

Knowledge Enhanced Methods have achieved remarkable success in Natural Language Processing, aiming to integrate external or prior knowledge to guide model prediction and enhance interpretability [11, 46, 49]. Following the proliferation of Large Language Models, techniques like Retrieval-Augmented Generation further showcase the efficacy of retrieving customized knowledge [15]. Recently, the strategy has been extended to Computer Vision, where external knowledge is leveraged to bridge the semantic gap between low-level features and high-level concepts. For instance, MLLM4WTAL [46] uses MLLMs to generate semantic priors for weakly supervised temporal action localization, and ENNGINE [49] employs a dual-branch framework for knowledge injection in incremental learning. Building upon the successful paradigms, our work applies this knowledge-centric viewpoint to event-based vision. We delve into the fundamental principles of event generation to explicitly mine the inherent polarity and motion knowledge within event streams, offering a new approach to enhance EAR.

3. Methodology

In this section, we introduce POKER, a new framework designed to enhance EAR by fully leveraging the inherent polarity-aware motion knowledge embedded in event streams. The overall architecture is shown in Fig. 2. We begin by presenting the preliminary concepts of event-based action recognition in Sec. 3.1. Building on the insights

introduced in Sec. 1, POKER leverages the intrinsic polarity information of events to enhance EAR. Specifically, Sec. 3.2 describes the proposed Polarity Motion Capturer, which explores the visual motion information encoded in event polarities. Then, Sec. 3.3 introduces the Polarity Motion Reasoner, which integrates a progressing polarity-aware prompt with the reasoning ability of a multimodal large model to generate textual semantic descriptions of polarity motion. Finally, Sec. 3.4 presents the overall training objective for incorporating this polarity-aware knowledge.

3.1. Event-based Action Recognition

Event cameras record brightness changes as a sequence of events $\mathbf{E} = \{e_i = (x_i, y_i, t_i, p_i)\}$, where each event e_i is denoted by its spatial coordinates (x_i, y_i), timestamp t_i, and polarity $p_i \in \{-1, 1\}$. To handle the asynchronous nature of $\mathbf{E}$, we first adopt [2, 52] to obtain original stacked or reconstructed event frames $\mathbf{I} \in \mathbb{R}^{T\times 3\times H\times W}$, where T, H, and W denote the temporal dimension, height, and width of the event frames. $\mathbf{I}$ are fed into the event encoder $\mathcal{E}_{\mathrm{V}}$ to encode event feature $\mathbf{F}_{\mathrm{V}} = \mathcal{E}_{\mathrm{V}}(\mathbf{I}) \in \mathbb{R}^{1\times D}$. The text encoder $\mathcal{E}_T$ [51] utilizes action label prompts: *"A series of photos recording human action of [cls]."* for each class to encode text features $\mathbf{F}_{\mathrm{T}} = \mathcal{E}_{\mathrm{T}}(\mathbf{T}) \in \mathbb{R}^{C\times D}$ as its classifier, where $\mathbf{T}$ represents all action label prompts, C denotes the number of classes. During inference, the final class probability $p \in \mathbb{R}^{1\times C}$ is computed on event and text features:

$$p = \mathrm{Softmax}(\mathbf{F}_{\mathrm{V}}(\mathbf{F}_{\mathrm{T}})^{\top}),\ c = \mathrm{argmax}(p), \tag{1}$$

where c represents the final action prediction class.

3.2. Polarity Motion Capturer (PMC)

The polarity in an event stream can be divided into positive and negative types. The positive and negative polarities capture complementary aspects of motion, both representing essential differential motion information, as shown in Fig. 3. However, existing methods often stack both polarities into a single event frame, which inevitably weakens polarity-specific cues during spatio-temporal encoding. To fully exploit this intrinsic information, we propose the **Polarity Motion Capturer (PMC)**, which explicitly separates the two polarities and captures their respective spatio-temporal dynamics using a correlation-based strategy, thereby enhancing the event encoder's ability to understand actions.

To capture the motion cues inherently embedded in event polarity, the PMC decouples the raw event stream $\mathbf{E}$ based on the polarity value p. This process yields two distinct streams: the positive polarity events stream $\mathbf{E}_{\mathrm{PP}} = \{e_i = (x_i, y_i, t_i, p_i) \mid p_i = +1\}$ and the negative polarity events stream $\mathbf{E}_{\mathrm{NP}} = \{e_i = (x_i, y_i, t_i, p_i) \mid p_i = -1\}$, which are stacked to form positive polarity frames $\mathbf{I}_{\mathrm{PP}}$ and negative polarity frames $\mathbf{I}_{\mathrm{NP}} \in \mathbb{R}^{T\times 3\times H\times W}$. Subsequently,

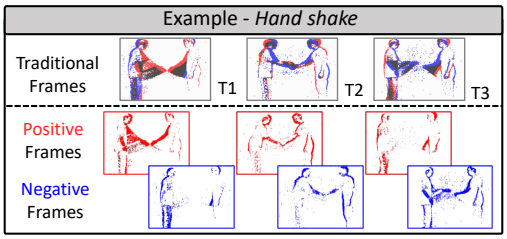


Figure 3. Example between traditional and decoupled frames.

PMC delves into the spatio-temporal correlations within these polarity frames in depth using the frame differencing method, including the *Intra-Polarity Temporal Correlation* of the same polarity and the *Inter-Polarity Spatial Correlation* between frames of different polarities.

Intra-Polarity Temporal Correlation. It captures the smoothness and continuity of motion within the same polarity by measuring the differences between consecutive time steps in the corresponding polarity frame flow:

$$\begin{aligned}\mathbf{I}_{\mathrm{PTC}}(t) &= \mathbf{I}_{\mathrm{PP}}(t) - \mathbf{I}_{\mathrm{PP}}(t-1),\\ \mathbf{I}_{\mathrm{NTC}}(t) &= \mathbf{I}_{\mathrm{NP}}(t) - \mathbf{I}_{\mathrm{NP}}(t-1),\end{aligned} \tag{2}$$

where $t \in \{1, \ldots, T\}$, $\mathbf{I}_{\mathrm{PTC}}(t)$ and $\mathbf{I}_{\mathrm{NTC}}(t)$ represent the intra-polarity temporal correlation of positive and negative polarities, respectively.

Inter-Polarity Spatial Correlation. It highlights the spatial differences between positive and negative polarities at the time step t, as $\mathbf{I}_{\mathrm{SC}}(t) = \mathbf{I}_{\mathrm{PP}}(t) - \mathbf{I}_{\mathrm{NP}}(t)$. These correlations $\mathcal{C} = \{\mathbf{I}_{\mathrm{PTC}}, \mathbf{I}_{\mathrm{NTC}}, \mathbf{I}_{\mathrm{SC}}\}$ constitute the core motion primitives extracted by PMC.

Dynamic Polarity Fusion. To dynamically adjust the contribution of each motion primitive $\mathbf{I}_m$ ($m \in \{\mathrm{PTC}, \mathrm{NTC}, \mathrm{SC}\}$), we propose a dynamic polarity fusion strategy. This strategy employs a gating function $\mathcal{G}(\cdot)$, implemented as a learnable routing matrix, to predict weighting coefficients based on the original event frame $\mathbf{I}$, enabling the model to prioritize polarity features ($\mathbf{I}_m$) that contribute most to the final discriminative representation. The polarity motion input ($\mathbf{I}_{\mathrm{P}}$) is the weighted sum of the correlation maps with $\mathbf{I}_m$.

$$\mathbf{I}_{\mathrm{P}} = \sum_m W_m \mathbf{I}_m, \tag{3}$$

where $\{W_m\} = \mathrm{Softmax}(\mathcal{G}(\mathbf{I}))$ is a vector representing the activation of each polarity predicted by $\mathcal{G}$.

3.3. Polarity Motion Reasoner (PMR)

The PMC produces polarity frames $\mathbf{I}_{\mathrm{P}}$ replete with motion information. Yet, it lacks a corresponding semantically aligned textual description, constraining its cross-modal potential. A potential solution is leveraging the strong visual localization and semantic reasoning abilities of multi-modal

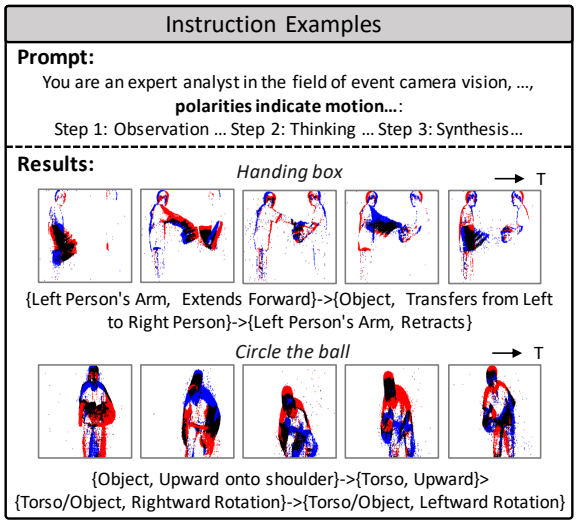


Figure 4. Instruction example for prompt and responses in PMR.

language models (MLLMs) to capture complex motion dynamics. However, MLLMs are not natively trained on raw event data, rendering direct application ineffective for fully utilizing their abstractive capacity. To overcome this limitation, we introduce the **Polarity Motion Reasoner (PMR)**, which strategically exploits the MLLM through polarity-specific guidance prompts tailored to motion characteristics. This design enables the MLLM to infer and summarize motion tendencies embedded in the event data, transforming visual dynamics into robust textual motion descriptions.

To fully exploit the reasoning capability of the MLLM, PMR introduces a progressive reasoning prompt $\mathcal{P}$ that guides the MLLM (ψ) to infer motion information from event frames in a step-by-step manner, producing an accurate motion description $\mathcal{T}$. The prompt $\mathcal{P}$ operates through three sequential steps.

Observation: In this initial step, ψ is instructed to localize all visible objects in each frame and recognize their instantaneous states (*e.g.*, standing or squatting in Fig. 4).

Thinking: Considering the temporal correlations inherent in event frames, ψ is guided to interpret the motion regions and trends within the current frame. It then refines this understanding by analyzing the motion relationships between the current and previous frames (*e.g.*, the positive polarity on the arm in the *"handing box"* action indicates the arm's forward movement trend, as shown in Fig. 4).

Synthesis: In the final step, ψ integrates the observations and reasoned insights to formulate a comprehensive motion description ($\mathcal{T}$) and predicts the action category. Given the difficulty of fine-grained action overlaps, ψ is explicitly constrained to ensure that the generated descriptions remain strictly grounded in visual evidence.

In practice, all steps are integrated into a single prompt for the MLLM, ensuring efficient inference. Examples of the PMR are shown in Fig. 4, and the complete prompt template is provided in *the Supplementary Material*.

3.4. Training Objectives

With PMC and PMR, the diverse polarity frames $\mathbf{I}_\mathrm{P}$ and the diverse motion descriptions $\mathcal{T}$ generated by MLLM bring intra-class variance within each action category, which makes alignment between them challenging. To address this issue, we introduce the **Polarity Alignment Loss** ($\mathcal{L}_\mathrm{P}$). This loss constructs a robust polarity class center by pulling the event polarity features $\mathbf{F}_{\mathrm{V,P}} = \mathcal{E}_\mathrm{V}(\mathbf{I}_\mathrm{P})$ and text polarity features $\mathbf{F}_{\mathrm{T,P}} = \mathcal{E}_\mathrm{T}(\mathcal{T})$ of the same class closer to their respective polarity centers, thereby reducing the adverse impact of intra-class variations.

Formally, the center vectors μ_V and μ_T are used for alignment,

$$\mathcal{L}_\mathrm{P} = \frac{1}{C}\sum_{c=1}^{C} \|\mu_\mathrm{V}^c - \mu_\mathrm{T}^c\|_2^2 . \tag{4}$$

where μ_V^c and $\mu_\mathrm{T}^c \in \mathbb{R}^D$ represent the mean feature vectors of the event polarity features $\mathbf{F}_{\mathrm{V,P}}^c$ and text polarity features $\mathbf{F}_{\mathrm{T,P}}^c$, respectively, for class c. With $\mathcal{L}_\mathrm{P}$, $\mathbf{F}_{\mathrm{V,P}}^c$ and $\mathbf{F}_{\mathrm{T,P}}^c$ achieve cross-modal alignment while maintaining tolerance to intra-class variations.

The task loss $\mathcal{L}_\mathrm{R}$ is constrained by the cross-modal alignment loss between the event feature $\mathbf{F}_\mathrm{V}$ and the text feature $\mathbf{F}_\mathrm{T}$ as follows,

$$\mathcal{L}_\mathrm{R} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\left(\mathbf{F}_{\mathrm{V},i}(\mathbf{F}_{\mathrm{T},i})^\top/\tau\right)}{\sum_{j=1}^{B}\exp\left(\mathbf{F}_{\mathrm{V},i}(\mathbf{F}_{\mathrm{T},j})^\top/\tau\right)}, \tag{5}$$

where τ is the temperature coefficient.

The final loss is composed of the task loss $\mathcal{L}_\mathrm{R}$, the polarity alignment loss $\mathcal{L}_\mathrm{P}$ as

$$\mathcal{L} = \mathcal{L}_\mathrm{R} + \alpha\mathcal{L}_\mathrm{P}, \tag{6}$$

where α is the balance coefficient between loss terms.

4. Experiments

4.1. Experiment Setups

Datasets. We evaluate our research on three benchmarks, including SeAct [52], DVS Action [21] and THU$^{\text{E-ACT}}$-50-CHL [7]. SeAct is a semantically rich dataset captured by a DAVIS346 event camera, containing 58 actions. DVS Action includes 450 recordings of 10 actions collected in office scenes with 15 subjects. THU$^{\text{E-ACT}}$-50-CHL includes 2,330 recordings from 18 participants captured in corridors and halls, lasting 2-5 seconds with diverse motion and illumination conditions.

Implementation Details. The implementation of the overall framework is carried out on PyTorch in a Linux environment with a single NVIDIA GeForce RTX 4090 GPU. We

Table 1. Comparative performance on the DVS Action, SeAct, and THU$^{\text{E-ACT}}$-50-CHL. The best results are in **bold**.

Method	Publication	Modality	Top-1 Accuracy (%)		
			SeAct	DVS Action	THU$^{\text{E-ACT}}$-50-CHL
HMAX SNN [40]	TNNLS'19	Event Stream	-	55.00	32.70
Motion SNN [18]	IJCAI'21	Event Stream	-	78.10	47.30
SpikePoint [24]	ICLR'24	Event Stream	-	90.60	-
TTPOINT [23]	MM'23	Event Stream	-	92.70	-
EventMamba [25]	TPAMI'25	Event Stream	-	89.10	59.40
EV-ACT [35]	TPAMI'23	Frame (stack)	-	92.60	58.50
SpikMamba [4]	AsiaMM'24	Frame (stack)	-	96.28	-
ESTF [35]	AAAI'24	Frame (stack)	-	-	49.50
PASS [50]	NeurIPS'25	Frame (stack)	66.38	-	-
ExACT [52]	CVPR'24	Frame (stack) + Text	67.24	94.83	-
EMP [2]	ICCV'25	Frame (recon) + Text	75.00	**99.80**	-
SAMPLE [30]	ICCV'25	Frame (stack) + Text	69.79	95.00	-
M2-CLIP [31]	AAAI'24	Frame (RGB) + Text	66.22	91.17	-
EZ-CLIP [1]	arXiv'23	Frame (RGB) + Text	67.98	92.95	-
EventBind [51]	ECCV'24	Frame (stack) + Text	67.24	94.73	60.77
+ POKER	-	Frame (stack) + Text	69.82 (+2.58)	96.49 (+1.76)	62.06 (+1.29)
EventBind [51]	ECCV'24	Frame (recon) + Text	74.13	98.24	61.14
+ POKER	-	Frame (recon) + Text	**76.72** (+2.59)	99.60 (+1.36)	**63.72** (+2.58)

Table 2. Ablation study for the effectiveness of each component in the proposed POKER.

PMC	PMR	SeAct (%)	THU$^{\text{E-ACT}}$-50-CHL (%)
✗	✗	74.13	61.14
✓	✗	75.00	62.24
✗	✓	75.86	62.98
✓	✓	**76.72**	**63.72**

use the Adam [14] optimizer with the initial learning rate of 1×10^{-5} and weight decay of 2×10^{-4}. CosineAnnealingLR learning rate schedule is employed with a minimum learning rate of 1×10^{-6}. Our model is trained for 100 epochs on DVS Action, SeAct, and THU$^{\text{E-ACT}}$-50-CHL.

4.2. Comparison with SOTA Methods

As shown in Tab. 1, we compare the proposed POKER with representative EAR methods across three benchmark datasets. The compared methods are grouped by input modality into: Event Point-based, Event Frame-based, and Event Frame/Text-based approaches. The Frame-based category is further divided into Reconstructed Frames and Stacked Frames. We also evaluate against newly introduced CLIP-based RGB video recognition methods, M2-CLIP [31] and EZ-CLIP [1]. From the results, it can be observed that Event Point-based methods generally yield sub-optimal performance, whereas Event Frame-based approaches achieve superior results owing to their denser spatial representation and stronger temporal consistency. Building upon the cross-modal EventBind [51] baseline, POKER incorporates polarity-aware motion enhancement and is evaluated under both reconstructed- and stacked-frame configurations. Across all datasets, POKER consistently improves the baseline performance, confirming its effectiveness in enhancing cross-modal alignment between events and text. Notably, on the highly challenging THU$^{\text{E-ACT}}$-50-CHL dataset, POKER achieves accuracy gains of 1.29% with stacked frames and 2.58% with reconstructed frames. These consistent improvements verify that the polarity knowledge enhancement effectively leverages the physical characteristics of event generation. By explicitly decoupling the positive and negative polarities, POKER captures the richer motion dynamics that is often suppressed during frame aggregation. This design enables the model to exploit the intrinsic motion cues embedded in event streams, thereby enabling more flexible and robust recognition across diverse datasets and input modalities.

4.3. Ablation Studies

Effectiveness of Each Component. As shown in Tab. 2, we evaluate the contributions of Polarity Motion Capturer (PMC) and Polarity Motion Reasoner (PMR) within the POKER on SeAct and THU$^{\text{E-ACT}}$-50-CHL datasets. The baseline is trained using the representative event encoder and text encoder from [51]. Building upon this baseline, incorporating PMC gains improvements of 0.87% on the SeAct dataset and 1.1% on the THU$^{\text{E-ACT}}$-50-CHL dataset, demonstrating that it is significantly effective for extracting

Table 3. Zero-shot recognition on the DVS Action dataset. "U" denotes understanding and "R" denotes recognition.

MLLM	Task	Top-1 Accuracy (%)
Qwen2.5-VL-72B	General-U	36.2
Qwen3-VL-30B	General-U	48.2
GPT-4o-mini	General-U	51.7
Gemini-2.5-pro	General-U	50.0
EventGPT	Event-U	6.0
EZSR	Event-R	84.7
Ours*	Event-R	**99.6**

Table 4. Different motion generation from PMR on SeAct dataset.

MLLM	Task	Top-1 Accuracy (%)
Qwen3-VL-30B	General	75.00
GPT-4o-mini	General	75.86
Gemini-2.5-pro	General	**76.72**

the crucial motion cues contained within the event streams. Notably, when PMR further imposes polarity-aware motion descriptions, there is a continuous improvement, namely improvements of 1.72% and 1.48% in Top-1 on the SeAct and THU$^{\text{E-ACT}}$-50-CHL datasets, respectively. These results validate the complementary roles of PMC and PMR, showing that their synergy effectively enhances polarity-guided motion understanding and overall recognition performance.

4.4. Diagnostic Studies

Impact of Different Motion Generation. As shown in Tab. 3, we first evaluate the zero-shot performance of several vision-language models [19, 45] on event-based action recognition, including both RGB-oriented MLLMs and the event-specific EventGPT [19]. The results reveal that all models struggle to generalize on event data due to its inherent sparsity and asynchronous characteristics. To address this limitation, we utilize the strong visual localization and motion reasoning capabilities of MLLMs to assist the Polarity Motion Reasoner (PMR), rather than relying on direct zero-shot inference. Tab. 4 further compares the effects of using different MLLMs within PMR, including GPT-4o-mini, Qwen3-VL-30B, and Gemini-2.5-Pro. Experiments on the SeACT dataset indicate that integrating MLLMs consistently enhances recognition performance, though the performance gain varies across models. Among them, Gemini-2.5-Pro achieves the best result, highlighting its stronger capability in extracting and reasoning about polarity-aware motion cues from event streams.

Impact of Different Fusion Strategy. As shown in Tab. 5, we compare three fusion strategies within PMC on the THU$^{\text{E-ACT}}$-50-CHL dataset, including simple concatenation, element-wise addition, and the proposed learnable routing-based fusion. The results indicate that the routing strategy achieves the best performance, demonstrating its ability to adaptively model inter-polarity dependencies and dynamically allocate information flow for more discriminative motion representation.

Table 5. Different fusion strategy on the THU$^{\text{E-ACT}}$-50-CHL dataset.

Method	Concat	Add	Router
Accuracy (%)	61.51	63.35	**63.72**

Table 6. Different $\mathcal{L}_{\text{P}}$ on the THU$^{\text{E-ACT}}$-50-CHL dataset.

Method	Contrastive Loss	Polarity Alignment Loss
Accuracy (%)	62.62	**63.72**

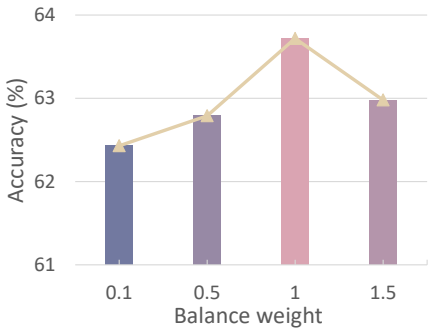


Figure 5. Effect of different values of α on THU$^{\text{E-ACT}}$-50-CHL.

Balance Between Alignment Learning. As shown in Fig. 5, maintaining an appropriate balance between the task loss $\mathcal{L}_{\text{R}}$ and the polarity alignment loss $\mathcal{L}_{\text{P}}$ is essential for stable and robust action learning (Eq. 6). The effect of $\mathcal{L}_{\text{P}}$, controlled by weighting factor α, reaches its optimum when $\alpha = 1$ on THU$^{\text{E-ACT}}$-50-CHL dataset. Furthermore, as reported in Tab. 6, replacing $\mathcal{L}_{\text{P}}$ with a standard contrastive loss leads to a performance drop from 63.72% to 62.62%. This result verifies that the proposed polarity alignment effectively improves both feature discriminability and overall recognition robustness.

4.5. Qualitative Analysis

Visualization of Textual Motion Descriptions. As illustrated in the Fig. 6, we visualize the textual motion descriptions generated under different instruction strategies to evaluate the effectiveness of the proposed PMR. In Fig 6 (a), using generic instructions, the MLLM performs a frame-by-frame motion analysis. However, this approach yields inaccurate or ambiguous descriptions when inter-class differences are subtle, as the resulting motion cues from dense or overlapping frames appear similar. In contrast, Fig. 6 (b) shows results from our polarity-guided prompting, which explicitly leverages the principle that opposite polarities encode distinct motion tendencies. This targeted guidance enables the MLLM to generate more discriminative and coher-

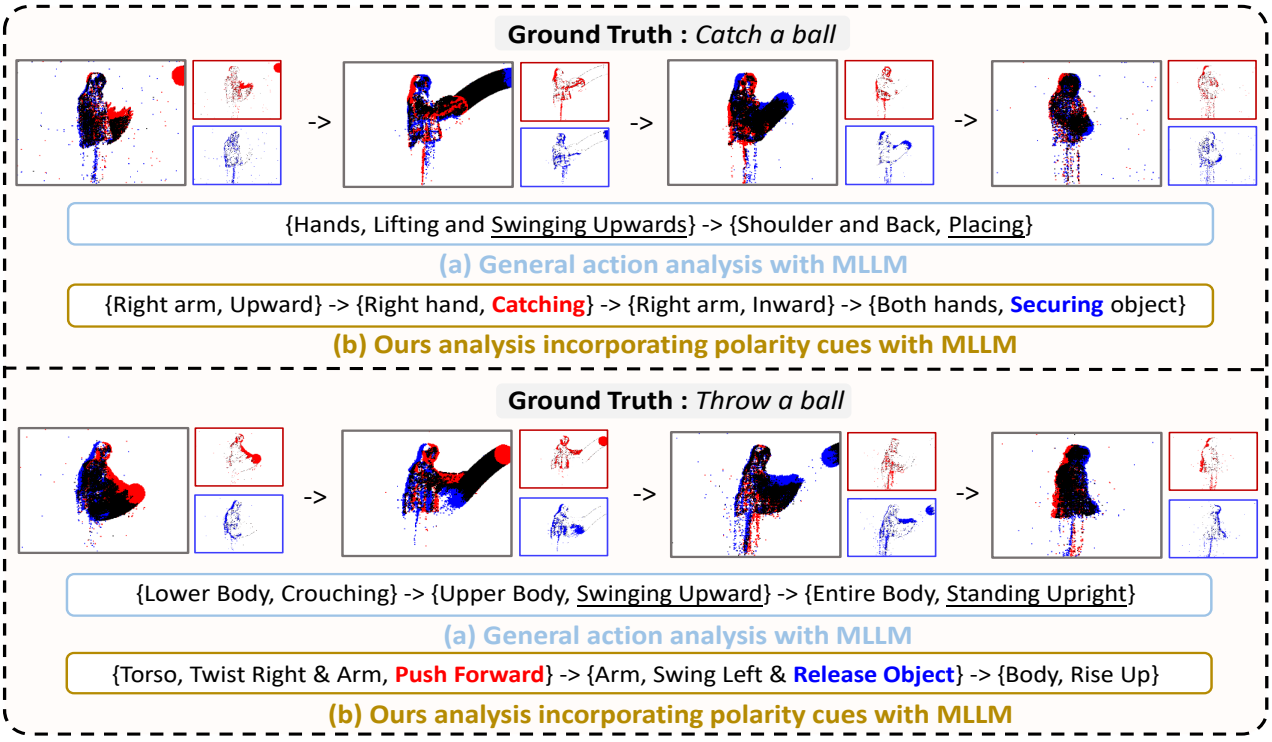


Figure 6. Comparison visualization of textual motion descriptions reasoned by MLLM. (a) Event-action analysis using generic instructions with MLLM; (b) Our analysis incorporating polarity motion cues inherent in the event with MLLM.

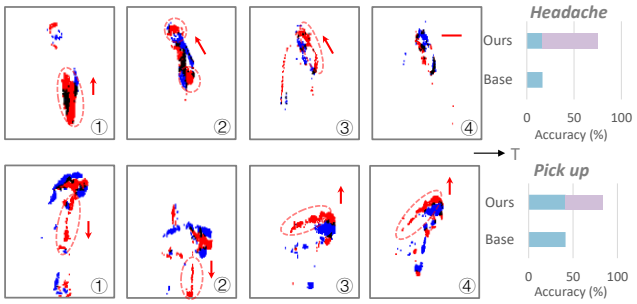


Figure 7. Comparison visualization of recognition on the challenging THU$^{\mathrm{E-ACT}}$-50-CHL dataset.

ent motion descriptions, clearly distinguishing actions such as "catch a ball" and "throw a ball".

Visualization of Recognition Results. To further investigate the effectiveness of the proposed polarity knowledge enhancement, we visualize the recognition results for two representative actions, "headache" and "pick up," from the highly challenging THU$^{\mathrm{E-ACT}}$-50-CHL dataset, as shown in Fig. 7. We compare the baseline EventBind [51] with our POKER framework. As observed, the inherent sparsity of event frames poses difficulties for direct recognition, often leading to ambiguous motion interpretation in the baseline model. In contrast, POKER effectively integrates dual-modality polarity knowledge, enhancing the perception of fine-grained motion tendencies and spatio-temporal transitions. These visual results confirm that the injected polarity-aware motion cues substantially improve the model's ability to capture discriminative action dynamics, achieving more precise and semantically coherent recognition.

5. Conclusion

In this work, we propose the POlarity Knowledge Enhanced framework (POKER) to address the critical overlook of intrinsic polarity-driven dynamics in existing cross-modal approaches. POKER establishes a novel cross-modal learning paradigm that explicitly enriches spatiotemporal event representations by integrating polarity-aware motion knowledge from both visual and textual domains. The framework consists of two synergistic components: the Polarity Motion Capturer (PMC), which effectively decouples visual polarities to explore their motion patterns thoroughly; and the Polarity Motion Reasoner (PMR), which semantically reasons polarity-induced motion variations via MLLMs. Through the polarity alignment mechanism, POKER successfully couples high-level semantic reasoning with low-level visual dynamics, leading to the acquisition of highly discriminative features. Extensive experiments across multiple benchmarks validate that POKER achieves superior performance.

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